Declaration of AI Use

A Behavior-Tree Framework for Channel-Aware Failure Recovery in Decentralized Multi-Robot Task Allocation under Bursty Communication

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A. AI used in the project

1. Permutation and combination analysis. I used AI to set up and simplify the combinatorial parts of my analysis, which were tedious and error-prone by hand. The clearest example is working out how many of the candidate robot meeting points were actually usable: for each pair of adjacent regions there were several hundred possible pairings of nearby survey points, and I needed to know how many met my distance and timing conditions. AI let me enumerate and filter these quickly, which is how I found that only one to three pairs in an entire instance qualified - a result that changed my design.

2. Checking for overlapping existing methods. I used AI to search the literature for methods similar to mine, specifically to establish whether my approach was already published and therefore not worth pursuing. This was decisive rather than a formality. It surfaced a 2026 *Computer Communications* paper covering substantially the same combination of ideas as my earlier method, which led me to set that method aside, and later a co-observation planning paper using the same missed-meeting detection signal I had proposed. In both cases I read the sources myself to confirm the overlap before acting on it.

3. Coding complex components of my simulation. While building my own simulation framework for testing, I used AI mainly for the HoloOcean integration. HoloOcean is a niche research simulator with a small user community and limited documentation or worked examples online, so there was little to work from beyond the API itself. I used AI for the scenario configuration, the vehicle control and ocean-current interfaces, and the experiment harness - in one instance a vehicle type behaved contrary to what the interface suggested, which only exploratory test code could establish. I also used it for some of the more intricate routines elsewhere in the code. I specified the intended behaviour throughout and verified outputs against test cases.

4. Identifying loopholes and testing novelty. Most importantly, I used AI as a critic of my own work - to look for weaknesses in my reasoning and to check whether my solution was genuinely novel or overlapped with existing work. This changed my conclusions more than once. In the most significant instance, a modelling assumption underpinning one of my main results was tested against independent physics simulation and shown to be false, and I withdrew the associated claims. In another, a comparison I had reported turned out to be an artefact of an assumption rather than a real effect, and I corrected it.

B. AI used in the drafting of the document

1. I used **Grammarly** to correct grammar and spelling.

2. I used **Claude** to refine my draft - improving word choice and sentence structure so the writing reads in an appropriately professional and technical register. The substance, structure and arguments are mine; this use was to improve how they were expressed.

3. I used **Claude** to generate LaTeX code for the more complex mathematical expressions in the paper, such as the multi-term variance expressions and the constrained optimisation statements, which are laborious to typeset correctly by hand.

I verified the outputs of all of the above against my own data and reasoning, and I take full responsibility for the content of this work.